Beamer$^+$

Interactive Presentations

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Beamer^{+} is a lightweight web app that enhances static slides with embedded media, real-time drawing tools, intuitive navigation, and a suite of powerful presentation features.

using Beamer^{+}

follow these steps to use Beamer^{+}

(1) **create**

create slides using any standard tool (including LaTeX)

(2) **configure**

write `config` files to specify the placement and behaviour of any embedded media

(3) **package**

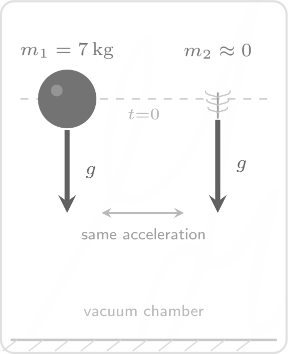
combine the slides, embedded media, and `config` files into a package

(4) **present**

open the package with Beamer^{+}, which will display the slides with all embedded media integrated seamlessly

videos

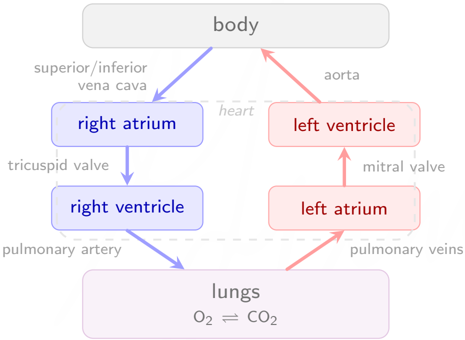
supported formats include `.mp4`, `.mov`, `.mpeg`



gravitational acceleration is
independent of mass

models

supported formats include `.glpb`



blood flow through the heart
(deoxygenated blood / oxygenated blood)

audio

supported formats include `.mp3`, `.wav`

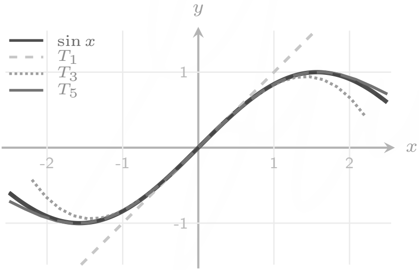
custom widgets

supported formats include `.html`

Any smooth function can be approximated by polynomials.

A function f can be approximated around $x_0 = 0$ as follows:

$$f(x) \; = \; \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}\, x^n$$



step	approximation
T_1	x
T_3	$x - x^3/3!$
T_5	$x - x^3/3! + x^5/5!$
T_5	$x - x^3/3! + x^5/5! - x^7/7!$
$\vdots$	$\vdots$

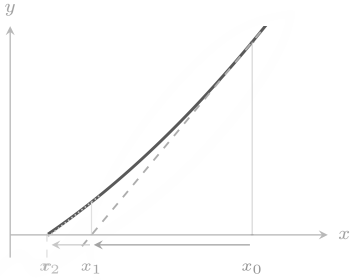
expansion of $f(x) = \sin(x)$ around $x_0 = 0$

Newton's method finds roots iteratively by following tangent lines.

Given $f(x) = 0$ with initial guess x_0, iterate:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Each step replaces f by its **tangent line** at x_n and takes the root of that line as the next guess.



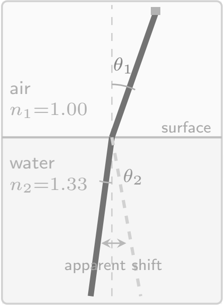
using Newton's method to solve $x^2 - 2 = 0$ from $x_0 = 2$ ($x = \sqrt{2}$)

Light bends when it crosses the boundary between two materials.

At the edge between two media, with refractive indices n_1 and n_2:

$$n_1 \sin\theta_1 \;=\; n_2 \sin\theta_2$$

where θ_1 is the angle of incidence, and θ_2 is the angle of refraction.



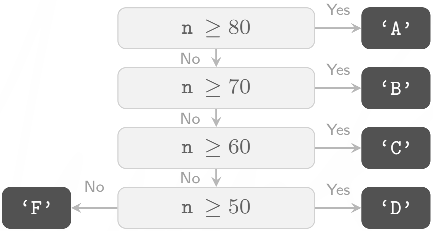
refraction of a pencil in a class of water

Complete the function, `letter_grade` that takes as input a student's numerical grade, `n`, and returns its corresponding letter grade:

numerical grade, `n`	letter grade
$n \geq 80$	A
$70 \leq n \leq 79$	B
$60 \leq n \leq 69$	C
$50 \leq n \leq 59$	D
$n \leq 49$	F

numerical to letter grade conversion

A flowchart for an algorithm to convert a numerical grade, `n`, to a letter grade is shown below:



numerical to letter grade conversion algorithm

learn more

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`